**UNIVERSITY OF TEHRAN**
**Electrical and Computer Engineering Department**
**Digital Logic Design, ECE 367 / Digital Systems I, ECE 894**
**Spring 1404**
**Computer Assignment 1**
**Basic Switch and Gate Structures in SystemVerilog**
**Week 3**

**Name:**

**Date:**

**1.** Generate switch level descriptions for a 3-input OAI (w = ~( (a | b) & c) ) as a complex gate, and a 2-input NOR in Verilog. Use #(3, 4, 5) delay for the NMOS transistors and #(5, 6, 7) for the PMOS transistors. Develop a testbench to test the two circuits for various input combinations. In the testbench test the circuits for their worst-case To1 and To0 delays. Make sure you distance your test data changes enough to allow all propagations to occur before changing the inputs.

**2.** Using the NOR gate Part 1, generate another version of OAI with separate gates. Develop a testbench to test this OAI for various input combinations. In the testbench, test the circuit for its worst-case To1 and To0 delays.

**3.** Write a description for the OAI of Part 1 using an **assign** statement. Use worst-case delay values from Part 1. Generate a testbench for this circuit and test it for various input combinations.

**4.** Generate a testbench for the OAI circuits of Part1, Part 2, and Part 3 and compare the functional responses and their timings. Explain the differences. In some cases, change the inputs before complete propagations occur and observe the outputs. Explain the responses.

**5.** Using three OAI gates build the function g() described below:

$j = \sim((a_1 + \sim b_1)\cdot\sim b_0)$

$k = \sim((\sim a_1 + b_1)\cdot 1)$

$g = \sim((\sim a_0 + j)\cdot k)$

**6.** Simulate the above circuit using delays you obtained in Part 1. If you treat $a_1a_0$ as a two-bit number, A, and $b_1b_0$ as B, what function is the g output of your circuit?

**Deliverables:**

Generate a report that includes item discussed below for each of the five parts of this CA. Do not show any code without first showing its circuit diagram.

A. Show the circuit diagram that you are analyzing. Show gates and/or transistors according to the specified delays.
B. Hand-simulate the circuits you have shown in Part A and write your expected values. For example, indicate what values you expect for the worst-case delays and express your reason for that.
C. Show your SystemVerilog descriptions of the design you are simulating and the testbench for it. Best is to use the Snip tools to get the image from Notepad++. This way you see all keywords and indentations. Make sure your SV codes are properly indented and all line-up rules are followed.
D. Show an image of the project that you have created in ModelSim for the simulation of your circuit.
E. When simulations are complete, or even if you have a partial simulation, include an image of the output waveforms showing signal names being displayed.

Make a PDF file of your report and name it with the format shown below:

*FirstinitialLastnameStudentnumber-CAn-ECEmmm*

Where *nn* is a two-digit number for the Computer Assignment, *mmm* is the three-digit course number under which you are registered, and hopefully you know the rest. For the *Firstinitial* use only one character. For *Lastname* and for the multi-part last names use the part you are most identified with. Use the last five digits of your student id (exclude 8101) for the *Studentnumber* field of the report file name.